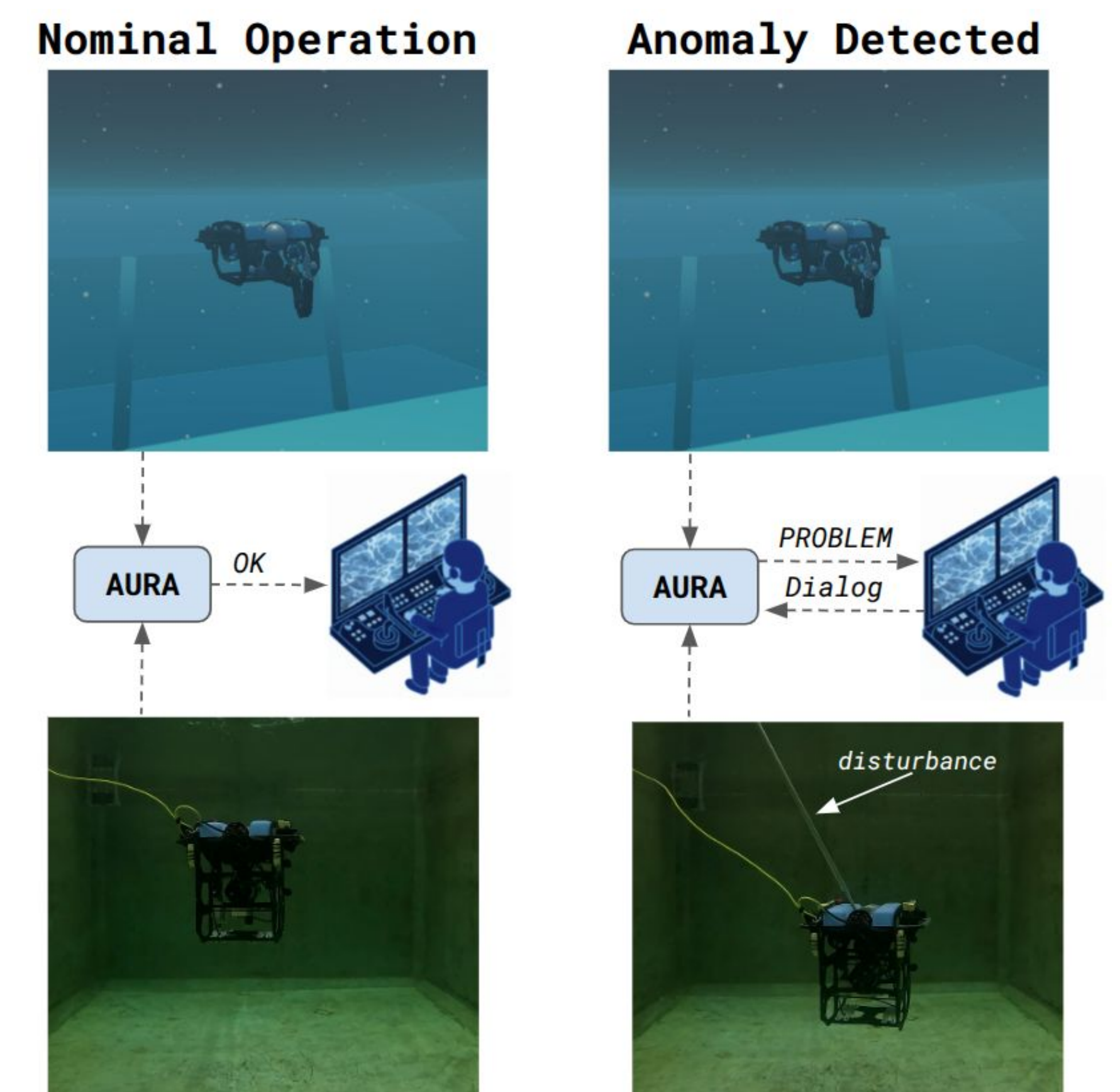


MOTIVATION

Traditional underwater robotics struggles with uncertainty and unexpected failures.

AURA (Autonomous Resilience Agent) combines a physics-based Digital Twin with dual AI agents and human oversight to diagnose unforeseen underwater anomalies in real time.



THE MAIN CONTRIBUTION

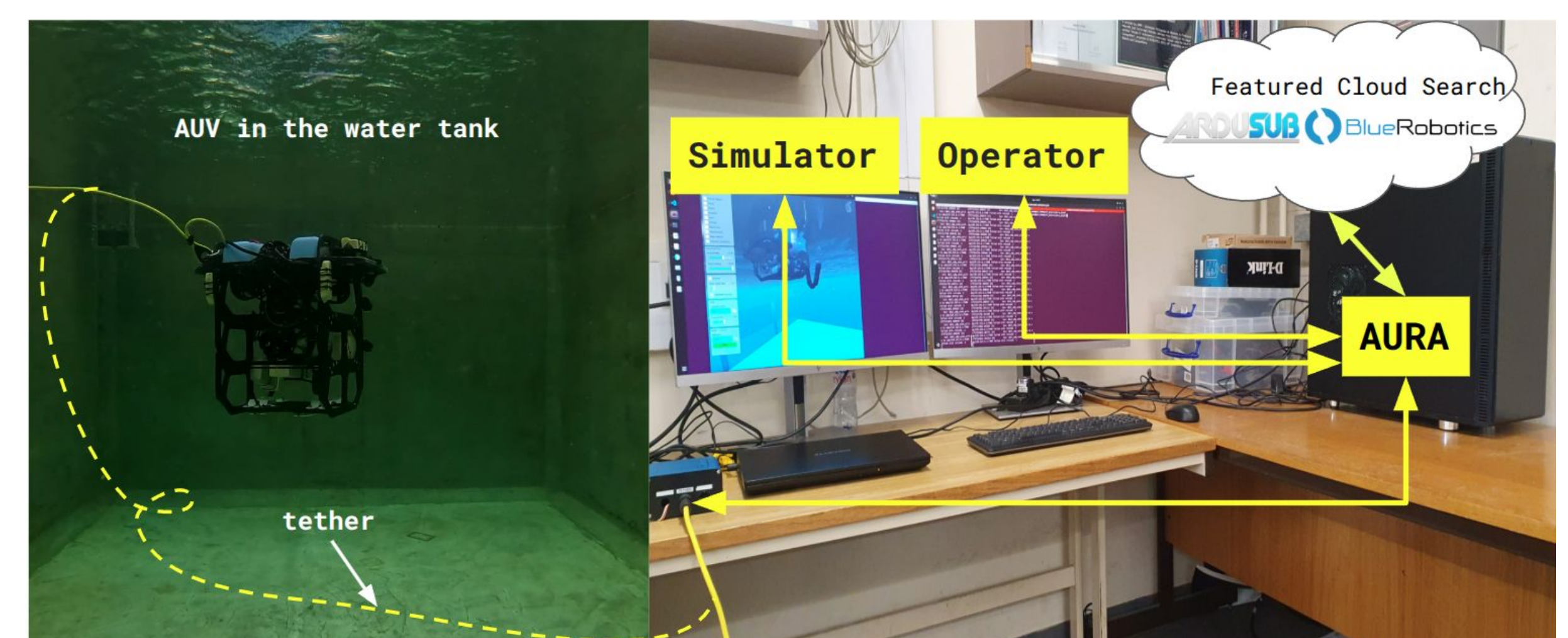
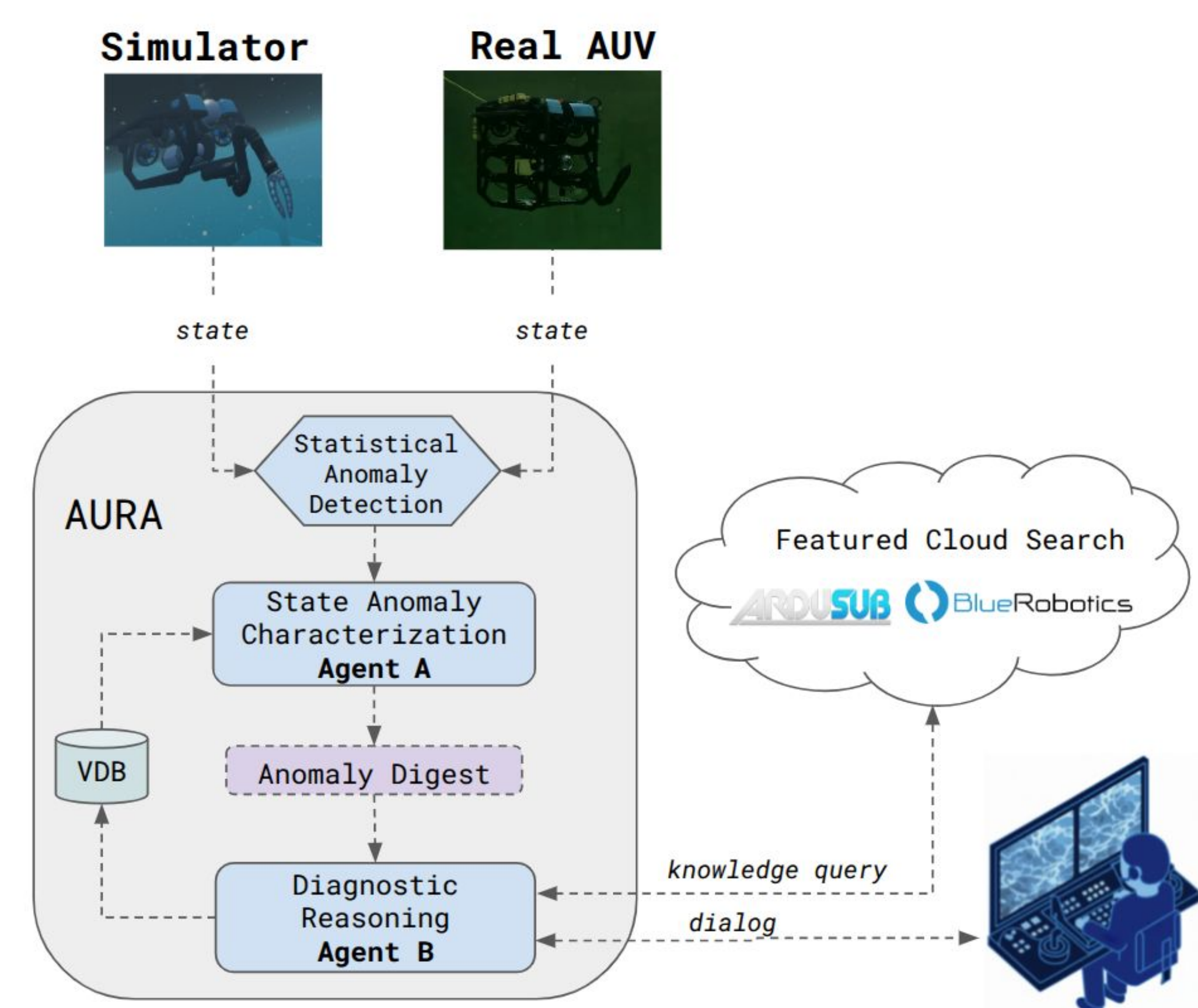
Two-Agent Architecture:

A low-level agent translates raw vehicle telemetry into natural-language problem descriptions, while a high-level reasoning agent interacts with the human operator to determine root causes.

Digital Twin Baseline: Parallel physics simulation in Stonefish monitors state residuals via Mahalanobis Distance while adapting to environmental drift.

Human-in-the-Loop Distillation:

Operator-validated diagnoses are distilled into an offline Vector Database to enhance future diagnostic precision without retraining.



KEY RESULTS

71% Fewer Dialogue Turns:

Diagnostic interaction reduced from 6.2 turns to 1.8 turns post-distillation.

4.8 / 5.0 Characterization Score

(CSS): Identifies the exact problem instead of guessing generic faults.

Few-Shot Efficiency: Required only 5 priming sessions to generalize across thruster, vertical, and rotational anomaly scenarios.

